This was independent research was carried out by the University of Bristol between December 2025 and April 2024. This document is a best attempt to capture the information gathered. Technology offerings change rapidly, and the authors take no responsibility or any decisions taken based on these findings, now or into the future.

If you are a heat pump manufacturer and would like to contribute to this work, please fill in this online form and we will get in touch: https://forms.office.com/e/n6YX8BrGhS

Heat Pump Remote Assistance Guide for Social Housing

Many heat pump providers now offer services that allow a resident's heat pump settings to be viewed and modified remotely. This is usually called "remote monitoring" however we prefer the term "remote assistance" as we feel this language is potentially less abrasive to residents/ tenants. We have surveyed a range of heat pump manufactures and third-party suppliers to understand how these services work and what advantages they could bring to housing providers. Not all providers offer the same functionality, and in this document we will try and highlight some of the features that you might want, and how to implement them.

The case for Remote Assistance

Remote visibility of heat pump settings provides a range of possible advantages:

1. Remote diagnostics can save engineers' time on site (BCC estimate cost of £100 per home visit on average)
2. Simple fixes can be performed remotely
3. System performance can be evaluated more easily
4. Usage patterns can be observed, enabling remote assistance viewers to support residents better

Fleet Management

Many heat pumps providers offer "dashboards", either online or via an app. This allows a housing provider's heat pumps to be viewed alongside one another, making it possible 'at a glance' to see if any are reporting faults. In some cases this "Fleet Management" functionality can also provide email notifications to a housing provider, informing them instantly of any faults that occur.

Smart Thermostats

As well as the remote assistance provided by the heat pump manufactures, there also exist a number of 3rd party providers that can use this remote-assistance data to improve the performance of the heat pump itself. This is generally referred to as "heat pump optimisation" but can be thought of as providing functionality not unlike a smart thermostat. It can also help reduce the commissioning setup burden on installers/ reduce the need for a revisit while also saving money for the tenant without impacting the level of comfort.

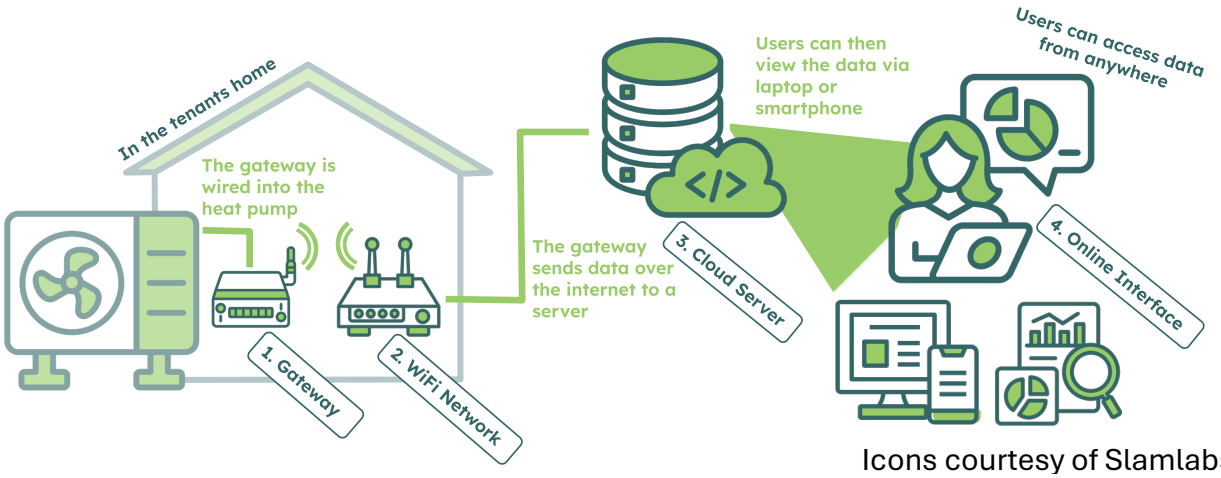
Comparison with Open Energy Monitor

This remote assistance functionality is different from the sort of heat pump monitoring provided by systems like the Open Energy Monitor (OEM). OEM provides very high-grade and independent visibility of heat pump performance, and does not rely on the stats being generated by the heat pump itself. OEM is read only however, and cannot be used to change setting. Generally this is a more robust but more expensive solution.
Remote assistance is different, with remote assistance you have visibility of heat pump stats and controls that originally you would have had to be present at the property to access. These stats, however, are produced by the heat pump, and are only as good as the heat pump's on board monitoring. Although no research grade testing across different brands has been made available publicly to compare manufacturer vs heat metered high accuracy monitoring, anecdotal evidence suggests a +- 10-20% margin of error.

How Remote Assistance Technology Works

Different manufactures do things in slightly different ways, but here we will try and give a high-level description of how the system works, and outline some of the terms that are used.

System Components



Icons courtesy of Slamlabs

1. *Gateway*

A heat pump must be fitted with a device that lets it communicate with the internet. These are manufacturer specific, and although some manufactures provide these by default, others do not and they must be specified separately. In this case there may be a cost associated with the purchasing of the gateway unit. More information is available in the comparison table given below.
Different manufactures call it different things, and it can be referred as a "gateway", a "smart controller", a "WiFi Interface" or a "WLAN Adapter" (amongst other things). Although this gateway is sometimes powered directly from the heat pump it sometimes requires an extra three pin socket to be available near the heat pump control unit.

2. *WiFi Network*

All of the manufactures we spoke to currently rely on WiFi to provide remote assistance capabilities. This is a limitation in the current technology. It either requires the tenant to have a reliable WiFi connection that they are willing to provide access to, or for the housing provider to install a separate WiFi network. Almost all the manufactures we spoke to are developing a SIM card-based alternative that would avoid need for WiFi, none yet have a product available. When these do become available they will simplify installation significantly, but may incur an operational cost. It is not possible to say exactly how high these costs will be, but with dedicated data SIMs it is possible to purchase 500MB of data for £9.00, and this should be more than enough to cover a year's worth of operation.
(For more information on how you might consider delivering a dedicated WiFi networks see section 7 of "Steps to Implementation" in this document.)

3. *Cloud*

The Gateway device gets data from the heat pump and sends it over the internet to computer servers, these servers are referred as the "cloud". Each manufacturer has a separate cloud. All of the manufactures we spoke to were aware of the data security and personal data requirements that this puts on them, and have located their servers appropriately. In most cases, heat pump data that is uploaded to the cloud is stored in a database so that it can be reviewed later to understand how a heat pump has been performing. In every case the server also hosts a website or presents information to users via an app.

4. *Online Interface*

As well as the in-home heat pump control unit, all manufactures provide some sort of online interface through which data can be viewed. In most cases there is both an app for use with a smart phone, and a website can be view of a laptop. The functionality of these interfaces varies slightly from manufacturer to manufacturer. All provide some visibility of a heat pump's current status, although not all provide error code.

Remote Assistance Features

The range of features provided by the remote assistance systems varies between manufactures. Available features include:

- Viewing and modifying settings such as heating set point. This allows a heat pump to be controlled remotely, as if one were at the in-house controller. This can also be used to diagnose and fix many of the common issues that arise from a miss configured heat pump, potentially reducing the number of site visits required.
- Viewing of live and historic heat pump performance data. This can include a wide range detailed features that would not necessary be available via the in-house unit. This can support the evaluation of heat pump performance and be used to identify where installers have not provided an adequate service. The values

available do vary between manufactures, with some not providing SCOP (or even COP) which can reduce the opportunities for system performance evaluation.

- Viewing and clearing of error codes. As well as telling a housing provider when a heat pump has stopped working, this feature can also support the maintenance engineer when diagnosing faults. This is often not hugely advanced however, and although error codes provide some guidance of what might be going wrong, site diagnosis is still required. One manufacturer's system is able to suggest which parts might be required to rectify an issue, but a detailed understanding of the heat pump in question is still required. Some systems enable the clearing of faults remotely, which could prevent some site visits to attend transient faults.
- Different user levels. Most systems provide the ability for different groups to have different levels of access. For example, some manufacturers provide a tenant's interface that doesn't have access to all the setting that the engineer's interface has.
- Email notification of faults. This means that engineers or housing officers are informed whenever there is a fault on a heat pump system, without having to continually check the dashboard.
- Application Interfaces (APIs). These allow data to be gathered from the server automatically. This is something that may be useful in future if a housing provider has a mixed fleet of heat pumps supplied by a range of manufacturers. API can be used to build a unified dashboard that displays all the different heat pumps. There are emerging offerings in this field to produce combined dashboards.

5. *Comparison between Manufactures*

To give an idea of the different features offered by different providers we have created a summary table. We will update this table as more suppliers approve information for dissemination. We are also creating an online databased, which we will also update as more providers input their product details.

Manufacturer	Grant
View Heating Set Point	✓
Modify Heating Set Point	✓

View Hot Water Temperature	✓
Modify Hot Water Temperature	✓
View Live System Data	✓
View Historic System Data	✓
Changing Heat Pump Setting	✓
View Error Logs	✓
Clear Errors	✓
Different User Levels	✓
Email Notifications	✓
APIs	✗
SIM cards	✗
Gateway requires separate power supply	✓

Current Limitation in the Technology Offerings

Although there are clear benefits to the adoptions of these technologies, there are some features that we would like to see become available to maximise their impact on the heat pump market. These include:

- The adoption of cellular (3G/4G/5G) wireless internet.
- More granular user access settings, with the opportunity to provide 3rd party advice organisations access to a user's system data in a controlled way.

- Logging of settings changes, with the ultimate aim being an integrated fault management and ticketing system that allows a housing provider to better manage and rectify faults.
- Standardised APIs, common across different manufactures, simplifying the creation of a unified dashboards in the future.

Smart Thermostats (Optimisation)

As well as the remote assistance services provided by the heat pump manufacturers there are also 3rd party providers. On top of offering the features described above these 3rd party providers also use machine learning to provide a range of more advanced control features designed to improve the heat pump system performance and reduce the cost of operation. This system improvement is generally referred to as smart thermostats functionality, or "heat pump optimisation". We must draw attention to the fact that we do not (as yet) have any experience of working with these systems, and the information provided below is from discussion with these 3rd party providers.
Different heat pump manufacturers work with different 3rd party smart thermostat systems, while some do not work with any.

How Smart Thermostats Work

As with the remote assistance more broadly, heat pump smart thermostats can work in a number of different ways, however there is commonality between them.
For smart thermostat functionality to be carried out data must be a sent to a different cloud sever, this time one that is run by the 3rd party provider. Sometimes data is sent there "as well as" to the heat pump manufacture, but sometimes it is sent "instead of", meaning that the manufactures remote assistance platform will no longer function and all remote assistance functionally must instead be carried out exclusively by the 3rd party smart thermostat provider.
Smart thermostat functionality uses machine learning to adjust the heat pump settings, with the aim of providing the desired comfort level for the minimum cost. The 3rd party smart thermostats use algorithms that look at the heat pump data over time and learn how the heat pump behaves. They will also learn the behaviour patterns of the tenants and develop an understanding of the thermal performance of the dwelling. The machine learning then combines this with data from other sources such as the weather forecast and (where relevant) upcoming changes in electricity price. Using all this data the machine learning will modify the heat pump settings, adjusting variables such as set point and flow temperature to try and provide the desired comfort level for the lowest possible price. Suppliers quote a range of potential savings, with 30% being the maximum saving we have seen quoted. We believe that further independent, real world studies should be undertaken to provide better accuracy of savings from these systems.

Advantages

As well as the potential savings, these systems have other potential advantages. These are:

- When using one of these smart thermostats there is potentially no need for an installer to fine tune a weather compensation curve, as the machine learning algorithm will select the most appropriate flow temperature at that given time (though a basic setup is expected first still). This has the potential to improve the quality of installations.
- Many smart thermostat suppliers work with a range of different heat pumps, and provide a unified dashboard through which all of the heat pumps they control can be viewed.
- Some smart thermostats have contracts with National Grid to provide "demand side management" services, which means that they are free to the user, and can actually provide some payment back to the customer.

Different Hardware/Software Approaches

These systems work in a range of different ways. In some case a unit must be installed in the home that connects directly to the heat pump via the Modbus interface. In this case this device must be provided with mains power and a WiFi connection, and often takes the place of the manufacturer's "gateway" (see "System Components" section above). Some of these systems do have the advantage that they are able to run offline for a period, and so if the internet is intermittent they can still provide some form smart thermostat functionality.
Some smart thermostat systems instead work "cloud to cloud". In this case there is no need for extra hardware in the home, as the smart thermostat provider simply takes the data directly from the heat pump manufacture's servers, and passes control signals back in the same way.

Steps to Implementation

In this section we will try and cover some of the steps that we suggest should be carried out when implementing remote assistance in a social housing setting.
These are just pointers, and can't be an exhaustive list of everything that must be carried out.

1. Supplier Selection

It is generally considered preferable to stick with a single or limited number of suppliers of heat pumps, as this reduces the number of different systems that maintenance and operations teams have to interact with. We recommend selecting manufactures that provide responsive and long term customer support. If you have existing heat pump stock then contact the manufacturer to identify if any remote assistance options are available for these models.

2. Specifying Heat Pump remote assistance in New Build and Retrofit Contracts

To guarantee that a specific manufacture of heat pump, with the required remote assistance capabilities is installed it is important to specify this in the house-building or retrofitting specifications within contracts that are let. Contracts teams might also want

to consider specifying the SCOP that a heat pump must achieve within a designated time frame if it is to be signed off.

3. Maintenance and Operational Team Upskilling

It is important that those dealing with the heat pumps on a daily basis are confident in their operation, and making the most of the remote assistance capabilities. Most heat-pump manufactures provide some sort of training and guidance for users, whether that be online or face to face.

A comprehensive hand over at the time of commissioning is also very important, focusing on how the heat pump system is controlled and how its performance is monitored. See other OptiHeat resources on this for further guidance.

Features and functionality are constantly changing, and heat pump manufacturers have all been interested in understanding the requirements of social housing landlords. To support this flow of information we recommend assigning a single individual within your organisation as a "Heat Pump Champion" who is in semi regular (every 4-6 months) contact with your selected heat pump manufacturer to see if there are any new heat pump and remote assistance features that might support operation.

4. Recruitment and Engagement

You are unlikely to get everything right on the first attempt of installing and commissioning these new systems. For this reason it is advisable to select responsive and understanding tenants for the first few installs, just while the people involved get to grip with the processes and requirements for the system. In general it should be noted that, without substantial engagement resource, more vulnerable tenants may not be suited to this technology role out within the early adopter phase.

Depending on the manufacturer the remote assistance technology can be transparent to the home owner, but some do have apps that allow remote control of the system, (similar to a smart thermostat would on a combi boiler), which might benefit some participants. Most of the smart thermostat systems are exclusively app-based, which might rule out the interaction of some tenants. Equally, a tenant is likely to have to sign a data privacy agreement, and allow the gateway to connect to their home WiFi. Therefore finding a participant that is willing to support the activity, at least while things get going, is important.

5. Data Protection and Privacy

Receiving data from homes requires the permission of the resident. This can be sought through an additional privacy notice or an updated tenancy agreement with the tenant. Manufacturers currently have Terms and Conditions that tenants sign up to at the time of system commissioning, but these are mostly designed for private owners and rarely cover the sharing of data with a housing authority.

The key elements to consider within this privacy notice are detail around who will have access to view system data and who will be able to change settings. This might simply be the heat pump manufacturer and the housing authority, but might also include 3rd party smart thermostat providers, 3rd party maintenance contractors, 3rd party advice providers (such as the Centre for Sustainable Energy) and others. The location of the

servers storing the data is another consideration, with any personal data being stored outside the UK requiring extra process.
A guiding principle whatever decision is chosen is that the tenants should always agree to the changes being made on the system and confirm this before any changes are made.
See the OptiHeat resources for an example privacy notice, although always consult with your internal legal and data teams before sending anything out.

6. Installation Requirements and Site Survey

There are things to look out for when surveying a site in preparation for installing heat-pump remote assistance. Specifically, some of the manufacturers require a separate power supply, in the form of the 3-pin plug, to be available. This is not always easy to provision, especially when retrofitting older properties with pre-existing heat pumps, and might require an electrician. We recommend where possible the fitting of a new, dedicated power supply on a fused spur to prevent the remote assistance gateway being accidently unplugged by the tenant.
If remote assistance is being retrofitted to an existing heat pump is it advisable to record the precise serial number of the heat pump in question, and confirming with the heat pump manufacturer that remote assistance is available. Sometimes it is also required to install an extra interfaces to the heat pump, and sometimes other element of the heat pump (such as the control board) may need to be replaced to enable the functionality. The same is true when retrofitting a smart thermostat system.

7. Provisioning Connectivity

As has been described, all of these systems require some form of internet connectivity if they are to provide remote assistance functionality. None of the manufacturers consulted to date currently provide an integrated cellular internet connection, although some did claim to very close to offering this (as of March 2026), and some do have “internet partners” that are able to provide a cellular connection through a secondary device.
The assumption from most manufacturers is that gateways will be connected to tenants WiFi, although in our experience this is not always a good approach, as tenant WiFi can be unreliable. This also complicates the data privacy agreement, and means that remote assistance cannot be installed before a tenant has moved into a property. Systems will report on a dashboard whether the internet connection is still active or whether the gateway has dropped off line. Checking remote assistance connectivity should become part of the regular maintenance process when visiting and inspecting a heat pump.

Conclusion

Hopefully we’ve demonstrated that remote assistance can be a useful tool for social housing providers, supporting them to in delivering a better heat pump experience to their tenants.

Our discussions with manufacturers have shown that there is an enthusiasm to realise this potential amongst the industry, although different manufacturers are at different points of the journey when it comes to delivering on this potential.

At the very least we would encourage you, when specifying a heat pump, to check with the manufacture what remote assistance they provide, although be aware that you as the housing provider may have to provide internet connectivity if the system is going to be reliable.